Supplementary Materials
HypEReg-TransMorph: A Hyperelastic-Regularized Transformer Framework for Folding-Suppressed Deformable Brain MRI Registration

This document accompanies the manuscript *HypEReg-TransMorph: A Hyperelastic-Regularized Transformer Framework for Folding-Suppressed Deformable Brain MRI Registration.* Extended quantitative results, inferential tables (S1–S2), grouped-VOI mapping (S3), Figure S1, OASIS protocol notes with Table S5, bootstrap confidence intervals (Table S6), Jacobian-ROI analysis placeholders (Table S7), LPBA40 zero-shot placeholders (Table S8), and implementation/profiling details are provided here as separate Supplementary Materials (they are not duplicated in the main journal PDF).

1 Supplementary Experimental Details

1.1 Model and Environment Configuration

To address reproducibility at full detail, we report *all* explicit training-time and implementation hyperparameters used by the IXI HypEReg-TransMorph run (`IXI/TransMorph/train_TransMorph_her.py` with `CONFIGS_TM['TransMorph']`).

Software and hardware environment.

- Python 3.12.10.
- PyTorch 2.12.0.dev20260408+cu128.
- antspyx 0.6.3; SimpleITK 2.5.3.
- Single GPU: NVIDIA RTX PRO 6000 Blackwell Workstation Edition (97,887 MiB VRAM), driver 591.86.

Data source, atlas resolution, and loaders.

- IXI root directory resolved as `repo_root/IXI_data`; atlas resolved from `atlas.pkl` (fallback `altas.pkl`).
- Training samples loaded from `IXI_data/Train/*.pkl`; validation samples from `IXI_data/Val/*.pkl`.
- Batch size: 2 (training), 1 (validation).
- DataLoader (train): `shuffle=True`, `num_workers=4`, `pin_memory=True`, `drop_last=False` (default).
- DataLoader (val): `shuffle=False`, `num_workers=4`, `pin_memory=True`, `drop_last=True`.
- Training transform pipeline: `RandomFlip(0)` then `NumpyType((np.float32, np.float32))`.

- Validation transform pipeline: `Seg_norm()` then `NumpyType((np.float32, np.int16))`.
- `RandomFlip` implementation detail: independent Bernoulli flips for axes $(1, 2, 3)$ with $\Pr(\text{flip along each axis}) = 0.5$.
- `Seg_norm` relabel table (in-order): $[0, 2, 3, 4, 5, 7, 8, 10, 11, 12, 13, 14, 15, 16, 17, 18, 24, 26, 28, 30, 31, 41, 42, 4$

Optimization, schedule, and training loop controls.

- Optimizer: Adam with AMSGrad (`optim.Adam(..., amsgrad=True)`).
- Adam hyperparameters explicitly set: learning rate $= 1 \times 10^{-4}$, weight decay $= 0$.
- Adam hyperparameters inherited from PyTorch defaults (not overridden): $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$.
- Epoch controls: `epoch_start=0`, `max_epoch=500`, `cont_training=False`.
- LR schedule (applied every training iteration): polynomial decay

$$\mathrm{lr}(e) = \mathrm{INIT_LR} \cdot \left(1 - \frac{e}{\mathrm{MAX_EPOCHES}}\right)^{0.9},$$

  with `round(..., 8)` applied in code.
- Bidirectional optimization per mini-batch: one step for $x \to y$, one step for $y \to x$, each with independent `zero_grad()`, `backward()`, `step()`.
- Random global seed: the released training script follows the upstream TransMorph reference implementation and does not pin a single global PyTorch/NumPy/Python seed; reproducibility is supported at the level of the released checkpoint, the deterministic evaluation pipeline, and per-case metric exports. A multi-seed extension (≥ 3 re-trainings) is identified in the main-paper limitations as a complementary direction.

Loss composition (all coefficients and internal settings). Total objective per direction:

$$\mathcal{L}_{\text{total}} = 1.0 \cdot \mathcal{L}_{\text{sim}} + 1.0 \cdot \mathcal{L}_{\text{grad}} + 1.0 \cdot \mathcal{L}_{\text{HypEReg}}.$$

- Similarity term: `losses.NCC_vxm(win=None)`.
- NCC window: $[9, 9, 9]$ (auto-selected from `win=None` for 3D).
- NCC return form: negative mean squared local cross-correlation ($-\overline{\mathrm{LCC}^2}$); denominator stabilizer $= 10^{-5}$.
- Smoothness term: `losses.Grad3d(penalty='l2', loss_mult=None)`.
- Grad3d form: mean squared first-order finite differences along x/y/z, then divided by 3.
- Hyperelastic term class: `HyperelasticLoss(alpha_length=0.0, beta_volume=0.02, gamma_fold=20.0)`.
- Hyperelastic internal defaults (not overridden): `eps=1e-3`, `fold_power=2`, `penalty_type='rational'`, `ablation='full'`.
- Active HypEReg weights in this operating point: $\alpha = 0$, $\beta = 0.02$, $\gamma = 20$.
- Jacobian computation in HER: forward finite differences on interior lattice $(D-1, H-1, W-1)$, with $J = \det(I + \nabla u)$.
- Volume penalty: $(J-1)^2 / \max(J, \epsilon)$ with $\epsilon = 10^{-3}$.
- Folding penalty: $\mathrm{ReLU}(\epsilon - J)^2$ with $\epsilon = 10^{-3}$.

Model architecture hyperparameters (TransMorph backbone). The run uses `config = CONFIGS_TM['TransMorph']` from `configs_TransMorph.py`, i.e.:

- `if_transskip=True`, `if_convskip=True`.
- `patch_size=4`, `in_chans=2`, `img_size=(160,192,224)`.
- `embed_dim=96`, `depths=(2,2,4,2)`, `num_heads=(4,4,8,8)`.
- `window_size=(5,6,7,7)` (as defined in the released config).
- `mlp_ratio=4`, `pat_merg_rf=4`, `reg_head_chan=16`.
- `qkv_bias=False`, `drop_rate=0`, `drop_path_rate=0.3`.
- Positional/normalization/checkpoint flags: `ape=False`, `spe=False`, `rpe=True`, `patch_norm=True`, `use_checkpoint=False`.
- Feature output stages: `out_indices=(0,1,2,3)`.
- Trainable parameter count noted in the config comment: 15,201,579.

Warping kernels, logging, and checkpoint policy.

- Two registration utility kernels instantiated during training: nearest-neighbor warp (labels) and bilinear warp (grid/intensity visualization).
- TensorBoard logging path: `logs/TransMorph_IXI_HER_ncc_1.0_grad_1.0_her_1.0_a0_b0.02_g20/`.
- Checkpoint path: `experiments/TransMorph_IXI_HER_ncc_1.0_grad_1.0_her_1.0_a0_b0.02_g20/`.
- Model-selection metric: best validation Dice (`best_dsc`).
- Checkpoint filename template: `dsc{:.3f}.pth.tar`; retention cap `max_model_num=8` with oldest-file deletion.

Inference/profiling controls (for reported runtime numbers).

- Inference benchmarks use the standardized evaluation script in the repository with harmonized metric definitions across models.
- Isolated forward profiling is performed with `torch.no_grad()` enabled, label-warp/metric kernels excluded, and `torch.cuda.max_memory_allocated()` reset before each warm-started measurement.
- Pure-forward runtime / peak GPU memory on the reported NVIDIA RTX PRO 6000 Blackwell hardware: HypEReg-TransMorph 0.0822 s / 5.685 GB; TransMorph 0.0822 s / 5.685 GB; TransMorphBayes 0.0852 s / 6.042 GB. Values are averages of 20 repeated calls on synthetic inputs of size $1 \times 2 \times 160 \times 192 \times 224$ following 5 warm-up iterations. The two TransMorph variants share the same forward graph and therefore yield identical timings (parameter count is 46.771 M and 46.773 M, respectively).

2 Figure S1: Descriptive IXI Overview

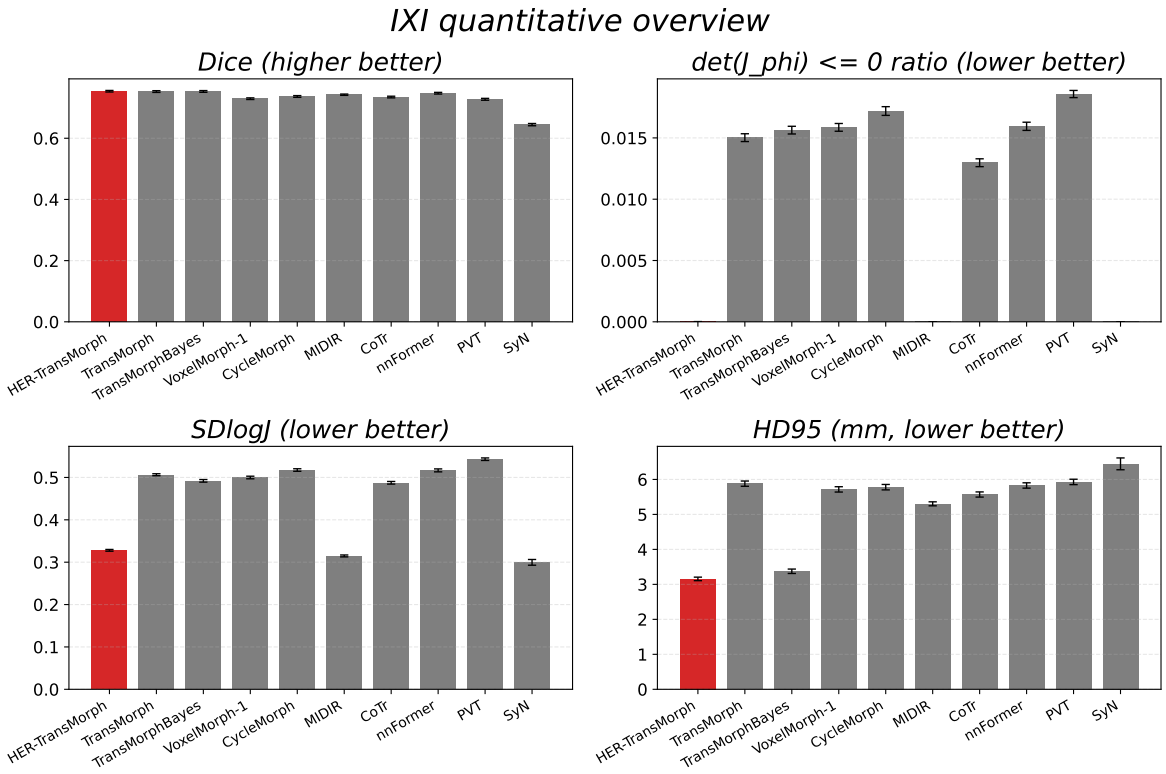

Figure S1: Descriptive IXI quantitative overview (Dice, non-positive Jacobian determinant ratio, SDlogJ, and HD95). Bars include standard-error error bars for visual uncertainty context.

3 Inferential Statistics (HypEReg-TransMorph vs. Baselines)

Table S1: Inferential summary for TransMorphBayes and MIDIR (mean±std, aligned median difference, Wilcoxon p, Benjamini–Hochberg q, and matched-pairs rank-biserial effect). HypEReg-TransMorph denotes the proposed model.

Metric	**Baseline**	**HypEReg mean±std**	**Baseline mean±std**	**Median (aligned)**	**diff**	p/q**/effect**
dice_mean	TransMorphBayes	0.5776±0.0303	0.5743±0.0315	0.00249		6.57×10^{-8} / 7.58×10^{-8} / 0.6522
non_jec	TransMorphBayes	$1.39 \times 10^{-5} \pm 7.55 \times 10^{-6}$	0.01484±0.00322	0.01449		1.19×10^{-16} / 1.99×10^{-16} / 1.0000
SDlogJ	TransMorphBayes	0.3268±0.0216	0.4908±0.0313	0.16264		1.19×10^{-16} / 1.99×10^{-16} / 1.0000
HD95_mean	TransMorphBayes	5.3032±0.7228	5.6881±0.7697	0.33877		1.34×10^{-14} / 1.88×10^{-14} / 0.9298
ASSD_mean	TransMorphBayes	1.3559±0.1809	1.4110±0.1839	0.05799		4.12×10^{-8} / 4.88×10^{-8} / 0.6622
dice_mean	MIDIR	0.5765±0.0301	0.5643±0.0244	0.01166		3.79×10^{-18} / 7.76×10^{-18} / 0.9331
non_jec	MIDIR	$1.46 \times 10^{-5} \pm 7.37 \times 10^{-6}$	0.0000±0.0000	-1.31×10^{-5}		1.31×10^{-20} / 3.69×10^{-20} / -1.0000
SDlogJ	MIDIR	0.3280±0.0221	0.3148±0.0242	-0.01365		4.84×10^{-20} / 1.28×10^{-19} / -0.9850
HD95_mean	MIDIR	5.3234±0.6936	5.3028±0.6139	-0.00855		0.5767 / 0.5898 / -0.0600
ASSD_mean	MIDIR	1.3570±0.1770	1.4100±0.1585	0.06514		3.58×10^{-9} / 4.36×10^{-9} / 0.6342

Aligned median difference is oriented so positive values indicate HypEReg-TransMorph improvement for the given metric direction. The effect is matched-pairs rank-biserial correlation. Pair counts differ by baseline (TransMorphBayes: $n = 91$; MIDIR: $n = 115$).

Table S2: Selected paired comparisons (Wilcoxon signed-rank with Benjamini–Hochberg FDR). Positive signed effect indicates HypEReg-TransMorph improvement under metric-aligned directionality; signed effect is matched-pairs rank-biserial correlation.

Metric	**Baseline**	**Median paired diff (aligned)**	**p-value**	**q-value**	**Signed effect**
non_jec	TransMorph	0.014747	1.31×10^{-20}	3.69×10^{-20}	1.0000
SDlogJ	TransMorph	0.170655	1.31×10^{-20}	3.69×10^{-20}	1.0000
HD95_mean	TransMorph	0.319583	3.77×10^{-17}	7.08×10^{-17}	0.9046
ASSD_mean	TransMorph	0.051747	6.09×10^{-11}	7.83×10^{-11}	0.7028
non_jec	MIDIR	-0.000013	1.31×10^{-20}	3.69×10^{-20}	-1.0000
SDlogJ	MIDIR	-0.013652	4.84×10^{-20}	1.28×10^{-19}	-0.9850
HD95_mean	MIDIR	-0.008554	0.5767	0.5898	-0.0600
ASSD_mean	MIDIR	0.065138	3.58×10^{-9}	4.36×10^{-9}	0.6342
non_jec	TransMorph (uploaded ckpt)	0.014747	1.31×10^{-20}	2.63×10^{-20}	1.0000
dice_mean	TransMorph (uploaded ckpt)	0.002326	6.88×10^{-7}	6.88×10^{-7}	0.5334
non_jec	TransMorphBayes (uploaded ckpt)	0.014714	1.31×10^{-20}	2.63×10^{-20}	1.0000
dice_mean	TransMorphBayes (uploaded ckpt)	0.002706	1.74×10^{-10}	2.32×10^{-10}	0.6858
non_jec	SyN	-0.000012	8.00×10^{-16}	1.24×10^{-15}	-0.8654
SDlogJ	SyN	-0.045599	3.09×10^{-5}	3.31×10^{-5}	-0.4477
HD95_mean	SyN	0.705730	1.31×10^{-7}	1.47×10^{-7}	0.5670
ASSD_mean	SyN	-0.111801	0.6572	0.6572	0.0477

Full inferential outputs across all exported metric-baseline pairs are provided in the supplementary release package. Additional uploaded-checkpoint paired results for TransMorph and TransMorphBayes (Dice and non-positive Jacobian ratio) are included with corresponding subject-level inputs. Repeated values such as $p = 1.31 \times 10^{-20}$ occur when the two-sided Wilcoxon test reaches the practical floating-point floor under near-complete sign consistency for $n = 115$, and should be interpreted as extremely small p-values.

4 Table S3: Grouped-VOI Mapping

Table S3: Grouped-VOI mapping used for grouped Dice reporting.

Grouped VOI	**FreeSurfer-compatible label IDs**
Brain-Stem	16
Thalamus	10, 49
Cerebellum-Cortex	8, 47
Cerebral-White-Matter	2, 41
Cerebellum-White-Matter	7, 46
Putamen	12, 51
VentralDC	28, 60
Pallidum	13, 52
Caudate	11, 50
Lateral-Ventricle	4, 43
Hippocampus	17, 53
3rd-Ventricle	14
4th-Ventricle	15
Amygdala	18, 54
Cerebral-Cortex	3, 42
CSF	24
choroid-plexus	31, 63

5 OASIS Cross-Cohort Training and Environment

The OASIS HypEReg-TransMorph run mirrors the IXI configuration in `OASIS/TransMorph/train_TransMorph_her.py` with `CONFIGS_TM['TransMorph']`, the same Adam/AMSGrad settings, polynomial LR decay, bidirectional updates, and HypEReg weights $(\alpha, \beta, \gamma, \epsilon) = (0, 0.02, 20, 10^{-3})$. Software versions should match the IXI block where possible (Python 3.12.x, PyTorch with a CUDA build appropriate to the host GPU, antspyx for SyN, SimpleITK for affine). Checkpoints are written under `OASIS/TransMorph/experiments/`; evaluation is driven by `OASIS/export_displacements.py`, `OASIS/eval_oasis.py`, `OASIS/run_full_eval.ps1`, and `OASIS/oasis_run_stats.py`.

6 Table S5: OASIS Paired Wilcoxon / BH-FDR

Reference model: `transmorph_her_oasis` (HypEReg-TransMorph, OASIS-retrained). All pairs n=19. Only significant contrasts ($q_{\mathrm{BH}} < 0.05$) shown; sign of direction: + = HypEReg-TransMorph better.

Table S4: Significant paired Wilcoxon contrasts (BH-FDR, $q < 0.05$) for HypEReg-TransMorph vs baselines on 19 OASIS test pairs.

Baseline	**Metric**	**HypEReg mean**	**Baseline mean**	p / q
CycleMorph	dice_mean↑	0.7966	0.7243	3.8×10^{-6} / 3.8×10^{-6}
	non_jec↓	1.0×10^{-5}	8.2×10^{-3}	3.8×10^{-6} / 3.8×10^{-6}
	SDlogJ↓	0.2786	0.4345	3.8×10^{-6} / 3.8×10^{-6}
	HD95_mean↓	2.294	3.076	3.8×10^{-6} / 3.8×10^{-6}
	ASSD_mean↓	0.680	0.949	3.8×10^{-6} / 3.8×10^{-6}
MIDIR	dice_mean↑	0.7966	0.7254	3.8×10^{-6} / 3.8×10^{-6}
	non_jec↓	1.0×10^{-5}	0.0000	1.3×10^{-4} / 1.3×10^{-4} (*HypEReg worse*)
	SDlogJ↓	0.2786	0.2551	3.8×10^{-6} / 3.8×10^{-6} (*HypEReg worse*)
	HD95_mean↓	2.294	2.893	3.8×10^{-6} / 3.8×10^{-6}
	ASSD_mean↓	0.680	0.931	3.8×10^{-6} / 3.8×10^{-6}
TransMorph (dsc0.857)	dice_mean↑	0.7966	0.8610	3.8×10^{-6} / 3.8×10^{-6} (*HypEReg worse*)
	non_jec↓	1.0×10^{-5}	8.1×10^{-3}	3.8×10^{-6} / 3.8×10^{-6}
	SDlogJ↓	0.2786	0.4211	3.8×10^{-6} / 3.8×10^{-6}
	HD95_mean↓	2.294	1.598	3.8×10^{-6} / 3.8×10^{-6} (*HypEReg worse*)
	ASSD_mean↓	0.680	0.483	3.8×10^{-6} / 3.8×10^{-6} (*HypEReg worse*)
VoxelMorph-1	dice_mean↑	0.7966	0.7159	3.8×10^{-6} / 3.8×10^{-6}
	non_jec↓	1.0×10^{-5}	8.1×10^{-3}	3.8×10^{-6} / 3.8×10^{-6}
	SDlogJ↓	0.2786	0.4291	3.8×10^{-6} / 3.8×10^{-6}
	HD95_mean↓	2.294	3.181	3.8×10^{-6} / 3.8×10^{-6}
	ASSD_mean↓	0.680	0.963	3.8×10^{-6} / 3.8×10^{-6}

p: two-sided Wilcoxon signed-rank; q: Benjamini–Hochberg FDR. "HypEReg worse" indicates the signed direction favours the baseline on that metric; these contrasts remain significant, indicating MIDIR/dsc0.857 significantly outperform HypEReg-TransMorph on those regularity/overlap dimensions. Full output in `OASIS/Eval_Results/_stats/oasis_pairwise_wilcoxon.csv`.

7 Table S6: Bootstrap 95% Confidence Intervals (IXI)

Table S5: Bootstrap 95% confidence intervals for IXI subject-level metric means ($B = 10{,}000$, sampling with replacement, seed = 0, $n = 115$ per model).

Model	**Dice mean [95% CI]**	**det(J_ϕ) ≤ 0 ratio mean [95% CI]**	**SDlogJ mean [95% CI]**	**HD95 mean [95% CI]**	**ASSD mean [95% CI]**
HypEReg-TransMorph	0.7537 [0.7486, 0.7587]	1.461×10^{-5} [1.332×10^{-5}, 1.604×10^{-5}]	0.3280 [0.3240, 0.3322]	5.3234 [5.1952, 5.4511]	1.3570 [1.3247, 1.3897]
TransMorph	0.7527 [0.7470, 0.7581]	1.502×10^{-2} [1.441×10^{-2}, 1.566×10^{-2}]	0.5008 [0.4943, 0.5076]	5.6872 [5.5391, 5.8376]	1.4073 [1.3735, 1.4429]
TransMorphBayes	0.7530 [0.7474, 0.7584]	1.510×10^{-2} [1.451×10^{-2}, 1.571×10^{-2}]	0.4934 [0.4874, 0.4996]	5.7246 [5.5847, 5.8646]	1.4160 [1.3831, 1.4501]
MIDIR	0.7423 [0.7380, 0.7465]	0 [0, 0]	0.3148 [0.3105, 0.3194]	5.3028 [5.1898, 5.4172]	1.4100 [1.3815, 1.4395]
SyN (ANTs)	0.6445 [0.6372, 0.6518]	5.874×10^{-6} [1.235×10^{-6}, 1.243×10^{-5}]	0.2996 [0.2866, 0.3130]	6.4457 [6.1174, 6.7784]	1.5233 [1.3838, 1.6675]

Dice follows the grouped 17-structure protocol in `IXI/analysis_trans.py`; non-positive Jacobian ratio, SDlogJ, HD95, and ASSD are computed from per-case exports in `IXI/Eval_Results`. Source file: `IXI/Results/comprehensive/bootstrap_ci_table_s6.csv`, generated by `scripts/bootstrap_ci.py`.

8 Table S7: OASIS Jacobian ROI Cleanliness Analysis

Placeholder for ROI-level Jacobian cleanliness statistics (hippocampus, lateral ventricles, cortical ribbon) and paired Wilcoxon/BH-FDR comparisons in the IXI→OASIS zero-shot setting. The analysis script is tracked in `scripts/oasis_roi_analysis.py`; results table will include fold ratio, median $\log\det J_\phi$, ROI SDlogJ, and adjusted q-values versus TransMorph.

9 Table S8: LPBA40 Zero-Shot Evaluation

Placeholder for LPBA40 zero-shot cross-cohort evaluation. Planned columns: Dice, non-positive Jacobian ratio, SDlogJ, HD95, and ASSD for HypEReg-TransMorph, TransMorph, TransMorphBayes, and MIDIR under identical IXI-trained checkpoints.